

Sachin Poudel, Ph.D. Candidate

✉ sachinscpdl@gmail.com | sachin.poudel@polsl.pl in sachinscpdl
🌐 <https://sachinscpdl.github.io/> 🎓 Google Scholar



Research Interest

Multiphysics Simulations for Functional Materials

- I am interested in multiphysics simulations of functional materials, including piezoelectric materials and batteries.
- My work focuses on coupled processes, such as microstructure evolution under external fields.
- I use phase-field modeling and physics-based simulations to understand how material structure affects properties and device performance.

Computational Modeling of Magnetic Materials

- I am also interested in computational modeling of magnetic materials.
- My focus includes magnetic domains, magnetization dynamics, anisotropy, magnetoelastic coupling, and phase transitions.
- I am interested in micromagnetic and multiscale simulations to study magnetic materials for spintronics, memory devices, sensors, and energy-efficient technologies.

Education

- Feb 2023 – Present 💡 **Ph.D. in Mechanical Engineering**, Silesian University of Technology, Poland.
Thesis: *Scale-Bridging Computational and Data-Driven Design of Microstructures in Multicomponent Alloys for Advanced Functional Materials*
- Aug 2020 – Sep 2022 💡 **M.Sc. in Mechanical Design and Manufacturing**, Tribhuvan University, IOE, Thapathali Campus, Nepal.
Thesis: *Towards Accelerated Design of Multi-Principal Element Alloys with Optimized Hardness and Elongation Features by an Ensemble of Neural Network*
- Nov 2014 – Oct 2018 💡 **B.E. in Mechanical Engineering**, Tribhuvan University, IOE, Thapathali Campus, Nepal.
Project: *Design, Fabrication, and Testing of Banana Fibre Extraction Machine*

Skills

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| Computational Methods | 💡 Phase-Field Modeling, Density Functional Theory, Finite Element Methods, Multi-Scale Simulations, Non-linear Electrochemical Modeling |
| Machine Learning | 💡 Data Preprocessing & Featurization, Neural Networks (TensorFlow/PyTorch), Material Informatics |
| Programming Languages | 💡 Python (NumPy, SciPy, scikit-learn, Streamlit), MATLAB, C++, Fortran, Data Visualization & Analysis |
| Software & Frameworks | 💡 MOOSE, Elmer, Quantum Espresso, ANSYS, Git, Linux |
| Languages | 💡 Strong reading, writing, and speaking competencies for English, Nepali |

Research Experience

Multiscale Computational Materials Design & Simulation

- ◆ Developed multiscale simulation frameworks linking DFT-based material parameters, phase-field modeling, continuum mechanics, and FEM device simulations.
- ◆ Implemented MOOSE-based phase-field models for Al–Cu–Ni microstructure evolution and lithium dendrite growth, including elastic, chemical, interfacial, and electrochemical effects.
- ◆ Applied finite element analysis and nonlinear solid mechanics, including finite-strain J_2 plasticity, to study deformation and electromechanical device performance.

Data-Driven Materials Discovery & Machine Learning

- ◆ Developed ML workflows for materials property prediction, inverse design, and optimization using feature engineering, PCA, correlation analysis, neural networks, and ensemble models.
- ◆ Built AlloyManufacturingNet for hardness–elongation synergy in multi-principal element alloys and PiezoTensorNet for crystal symmetry classification and full piezoelectric tensor prediction.
- ◆ Designed physics-guided ML and latent-space modeling approaches using literature-mined, experimental, and simulation data for piezoelectric nanofibers and lithium dendrite microstructures.

Functional Materials for Energy & Electromechanical Applications

- ◆ Studied piezoelectric materials, battery interfaces, multicomponent alloys, and additively manufactured stainless steels for energy and electromechanical applications.
- ◆ Optimized ceramics and polymeric piezoelectric systems by combining tensor prediction, crystallographic analysis, FEM simulations, and physics-guided ML.
- ◆ Connected process, structure, properties, and device performance through computational and data-driven modeling of functional materials.

Research Publications

- [1] S. Poudel, W. Smok, R. Thapa, A. Timofiejczuk, N. Moelans, and A. Kunwar, “Physics-guided machine learning for enhanced piezoelectric performance in electrospun pvdF/snO₂ nanofibers,” *Sustainable Materials and Technologies*, e02008, 2026. DOI: <https://doi.org/10.1016/j.susmat.2026.e02008>.
- [2] H. Tang, S. Poudel, M. Roszak, K. Vanmeensel, A. Kunwar, and N. Moelans, “Latent space exploration to decode the dynamic behavior of interfacial dendrites in lithium-ion batteries,” *Manuscript under review in Materials Reports: Energy, Elsevier*, 2026.
- [3] M. Dagnaw et al., “Heat treatment pathways and microstructural features determine the corrosion behavior of additively manufactured sdss 2507,” *Manuscript under review in Metals and Materials International*, 2026. DOI: [10.26434/chemrxiv-2025-sk6h5](https://doi.org/10.26434/chemrxiv-2025-sk6h5).
- [4] Y. Li, S. Poudel, S. Kumari, E. Azqadan, A. Kunwar, and N. Moelans, “Roadmaps for dendrite suppression in next generation lithium-ion batteries: Toward sustainable energy solutions,” *Energy Storage Materials*, p. 105 000, 2026. DOI: [10.1016/j.ensm.2026.105000](https://doi.org/10.1016/j.ensm.2026.105000).
- [5] S. Neupane, S. Poudel, A. Ghimire, and S. Shakya, “Enhanced mechanical performance and biodegradability in filler-reinforced potato biopolymers for sustainable packaging,” *Materials Today Communications*, p. 114 081, 2025. DOI: [10.1016/j.mtcomm.2025.114081](https://doi.org/10.1016/j.mtcomm.2025.114081).
- [6] S. Poudel, N. Moelans, R. Thapa, A. Timofiejczuk, D. Panthi, and A. Kunwar, “Unraveling elastochemical effects in microstructural evolution of al–cu–ni system through dft-informed multi-phase field simulations,” *International Journal of Solids and Structures*, vol. 300, p. 112 894, 2024. DOI: [10.1016/j.ijsolstr.2024.112894](https://doi.org/10.1016/j.ijsolstr.2024.112894).

- [7] S. Poudel et al., "Alloymanufacturingnet for discovery and design of hardness-elongation synergy in multi-principal element alloys," *Engineering Applications of Artificial Intelligence*, vol. 132, p. 107902, 2024. DOI: [10.1016/j.engappai.2024.107902](https://doi.org/10.1016/j.engappai.2024.107902).
- [8] S. Poudel, R. Thapa, R. Basnet, A. Timofiejczuk, and A. Kunwar, "Piezotensornet: Crystallography informed multi-scale hierarchical machine learning model for rapid piezoelectric performance finetuning," *Applied Energy*, vol. 361, p. 122901, 2024. DOI: [10.1016/j.apenergy.2024.122901](https://doi.org/10.1016/j.apenergy.2024.122901).
- [9] R. Thapa, S. Poudel, K. Krukiewicz, and A. Kunwar, "A topical review on ai-interlinked biodomain sensors for multi-purpose applications," *Measurement*, vol. 227, p. 114123, 2024. DOI: [10.1016/j.measurement.2024.114123](https://doi.org/10.1016/j.measurement.2024.114123).
- [10] U. Subedi, S. Poudel, K. Gyanwali, Y. Amorim Coutinho, G. Matula, and A. Kunwar, "State-of-the-art review on the aspects of martensitic alloys studied via machine learning," *Metals*, vol. 12, no. 11, p. 1884, 2022. DOI: [10.3390/met12111884](https://doi.org/10.3390/met12111884).
- [11] S. Poudel, S. Chapai, R. K. Subedi, T. R. Giri, and S. Adhikari, "Design, fabrication, and testing of banana fibre extraction machine," *Journal of Innovations in Engineering Education*, vol. 2, no. 1, pp. 165–173, 2019. DOI: [10.3126/jiee.v2i1.36668](https://doi.org/10.3126/jiee.v2i1.36668).

Conference Presentations

- 19 – 23 May 2024 ■ The 5th International Symposium on Phase-Field Modelling in Materials Science, Zhejiang University, Hangzhou, China.
"A computational model for examining the role of externally applied stress on dendrite growth pattern in solid state Lithium-ion batteries". (Oral Presentation)
- 3 – 7 Sep 2023 ■ Euromat23, Frankfurt, Germany "Phase field evolution in Ni-YSZ Fuel Cells with Temperature-Dependent Elasticity". (Oral Presentation)
- 19 – 22 Oct 2022 ■ 12th IOE Graduate Conference, Thapathali Campus, Nepal "Towards Accelerated Design of Multi-Principal Element Alloys with Optimized Hardness and Elongation Features by an Ensemble of Neural Network [ID: 12308]". (Oral Presentation)
- 05-06 April 2019 ■ Fifth Graduate Conference- Himalayan Knowledge Conclave 2019, Kathmandu University, Nepal. "Design, Fabrication and Testing of Banana Fibre Extraction Machine". (Poster Presentation)

Employment History

- April 2023 – Present ■ **Graduate Researcher**, Silesian University of Technology.
Research team member in the National Science Centre (NCN), Poland, funded project (UMO-2021/42/E/ST5/00339).
- Feb 2022 – Jan 2023 ■ **Lecturer**, COSMOS College of Management and Technology, Nepal.
Delivered lectures and instructed in laboratory on mechanical engineering subjects: Thermal Science (MEC 111) and Engineering Drawing (MEC 120).
- Nov 2021 – Feb 2023 ■ **Teaching Assistant**, IOE, Thapathali Campus, Tribhuvan University, Nepal.
Teaching core mechanical engineering courses: Engineering Drawing I (ME401), Engineering Drawing II (ME451), Computer Aided Drawing (ME 505).
- Jun 2019 – Nov 2021 ■ **Mechanical Design Engineer**, Calcgen Solutions, Nepal.
Conducted finite element analysis and stress analysis of pressure vessel components. Performed mechanical calculations and drafting of pressure vessels based on ASME Section VIII.
- Sep 2018 – Dec 2018 ■ **Intern**, Nepal Airlines Corporation, Nepal.
Diagnosed aircraft vehicles. Assisted in the repair and maintenance of various aircraft models.

Professional Achievements and Engagements

Awards and Recognitions

- 2024  **Rector's Grant, Silesian University of Technology, Poland.** Pro-quality competition for rector's grants for highly scored publications, granted patents, acquired projects, or scientific and research work
- 2023  **University Grants Commission, Nepal.** University Grants Commission (UGC) Master Research Support Award (Award No: MRS-78-79-Engg-10)

Workshops and Events

- 2 Jan – 5 Jan 2019  **Swiss-Nepal Technology Transfer Workshop 2019.** Organized by the Nepalese Scientific Association in Switzerland (NepSas)
- 4 Feb – 6 Feb 2016  **MechTRIX 7.0.** 7th National Mechanical Engineering Exhibition 2073